

ENGINEERING PORTFOLIO

Jacob Tenorio

Mechanical Engineering B.S. focusing on Aerospace Engineering

Washington University in St. Louis

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About Me

I am the Lead Structures & Test Engineer for WashU Design/Build/Fly. Through this club, I have cultivated an interest in aeronautical engineering. Through coursework in fluid mechanics, heat transfer, and thermal systems design, I have developed a parallel interest in thermal-fluid sciences. My goal is a career in a discipline such as turbomachinery, hot-section engineering, environmental control systems, or another branch of thermal management.

This interest dates back to my youth, when I helped my father with HVAC installations on weekend jobs. As I grew older, I began to wonder how air-conditioning units actually worked, which drew me toward power transfer; what I later learned in university was really thermal energy transfer. This intersection of interest in aircraft and thermal-fluid sciences has shaped an engineering career path I am eager to pursue.

As the Structures and Test lead, I oversee smaller sub-teams of students, each responsible for components critical to that year's competition missions. This has included addressing challenges like transportation constraints, which led one aircraft to feature removable wings. After a crash at a competition event, I expanded what had been a purely Structures team into Structures & Testing, introducing a greater emphasis on real-world validation than the team had previously used. This took the form of physical test rigs, used both to validate simulation results and to run niche experiments that are difficult to simulate. Competing under limited resources, especially in the aftermath of a crash, has sharpened my ability to think on my feet, identify weaknesses in existing methodologies, and improve the effectiveness of the teams I am part of and go on to lead. I weigh every idea my team brings to me, and I encourage my teammates to generate plenty of ideas, even ones that go unused, since the strongest solution tends to emerge from having many to choose from.

Washington University in St. Louis

Expected May 2027

B.S. in Mechanical Engineering **GPA: 3.51**

Relevant Coursework: Fluid Mechanics, Solid Mechanics, Heat Transfer, Thermal Systems Design, Thermodynamics, Machine Elements, Numerical Methods

PROJECT

Multi-Mission Competition Aircraft

AIAA Design/Build/Fly

SAE Aero Design

SolidWorks

Ansys

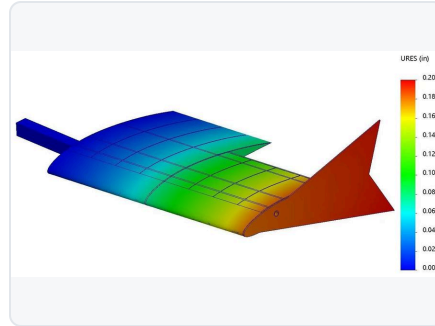
CFD

FEA

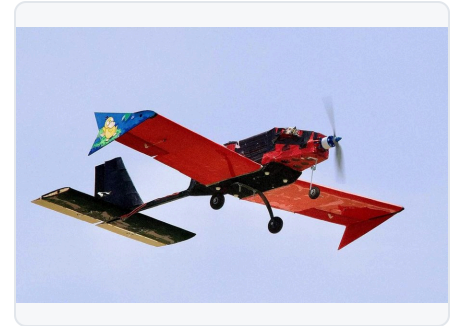
AIAA Design/Build/Fly Design Report – Primary Author (2026), Contributing Author (2025) – Detail Design Section



CAD MODEL



CFD ANALYSIS



FLIGHT PHOTO

WHAT?

- Designed aircraft for deployment missions (gliders/banners) and heavy cargo (passengers/pylons)
- Balanced high-lift capacity with low structural weight
- Lead structures and testing teams designing competition-compliant aircraft via SolidWorks top-down modeling

HOW?

- Modeled aircraft in **SolidWorks** using top-down modeling
- Integrated carbon fiber (spars), balsa (ribs), **3D-printed** PLA (complex mounts), and **CNC** machined parts (wingbox)
- Validated structural integrity (**FEA**) and aerodynamic performance (**CFD**)
- Designed mission-specific components (removable wings, deployable glider, cargo & passenger compartments)

RESULTS

- Placed **35th of 114** in international AIAA competition
- Achieved **100%** structural success rate with no components failing during flight
- Reduced new-member CAD training time by **50%** by designing an onboarding curriculum

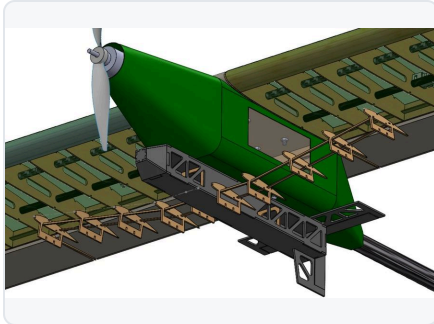
Glider Deployment Mechanism

Aerospace

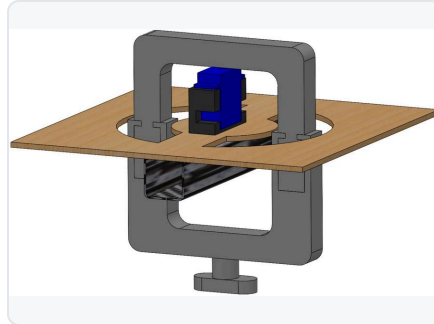
SolidWorks

FEA

3D Printing



CAD ASSEMBLY



DESIGN ITERATION



3D PRINT

WHAT?

- Designed lightweight release mechanism for mid-flight glider deployment
- Optimized for small-scale geometry and minimal weight impact on parent aircraft

HOW?

- Developed multiple **SolidWorks** designs, utilizing **3D-printed** PLA to test and refine release efficiency
- Engineered a one-step release mechanism to eliminate mechanical **failure points** and ensure mission reliability
- Performed stress analysis via SolidWorks **FEA** to identify critical points

RESULTS

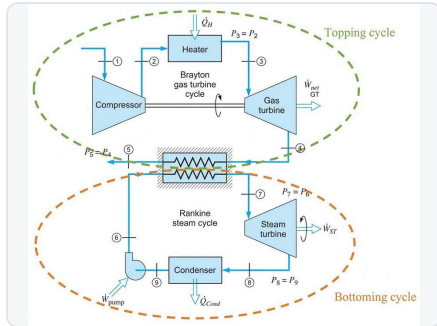
- Achieved **jam-free release** during Design/Build/Fly competition
- Simplified design proved effective under real-world flight vibrations and loads

CCGT Marine Propulsion System Design

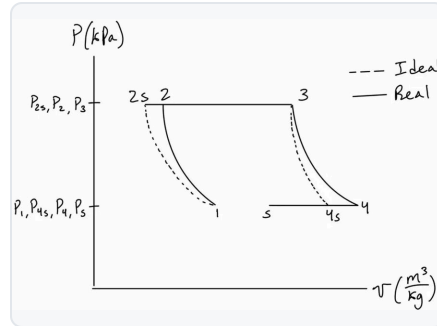
Thermal Systems

MATLAB

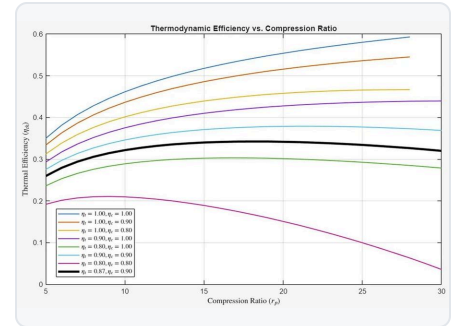
Python



BRAYTON CYCLE



RANKINE CYCLE



HRSG DESIGN

WHAT?

- Designed a combined-cycle gas turbine marine propulsion system
- Optimized both Brayton and Rankine cycles for maximum efficiency

HOW?

- Optimized variable-specific-heat Brayton cycle in **MATLAB** via parametric compression ratio and isentropic efficiency sweeps
- Maximized Rankine efficiency in **Python** via turbine pressure optimization with blade-erosion quality constraint
- Sized three-stage HRSG via Dittus-Boelter, Zukauskas, Rohsenow correlations, LMTD, and thermal resistance networks

RESULTS

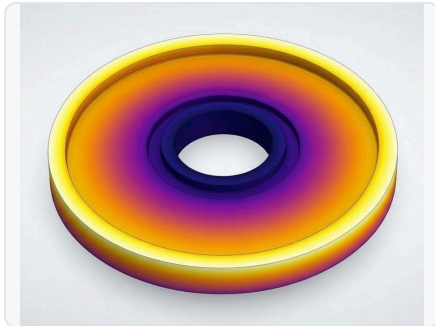
- Delivered optimized combined-cycle system meeting all design constraints
- Balanced thermodynamic efficiency with practical engineering constraints (blade erosion, quality limits)

Radial Heat Conduction in a Micro-Turbine Disk

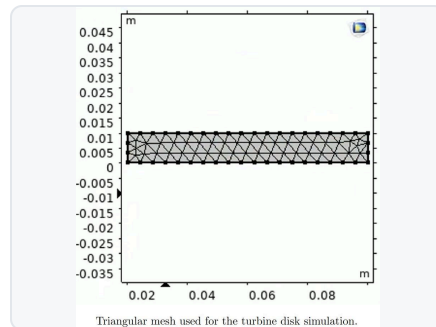
Heat Transfer

COMSOL

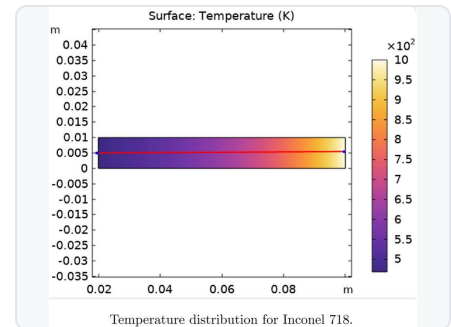
FEA



COMSOL MODEL



TEMPERATURE PROFILE



PARAMETRIC STUDY

WHAT?

- Simulated steady-state radial conduction in a micro-turbine disk
- Evaluated superalloy materials for turbine shaft thermal protection

HOW?

- Built **COMSOL** axisymmetric model with fixed rim temperature, forced convection, and lubricant cooling BCs
- Validated against Bessel fin analytical solution
- Performed parametric study and mesh independence testing across 3 superalloys

RESULTS

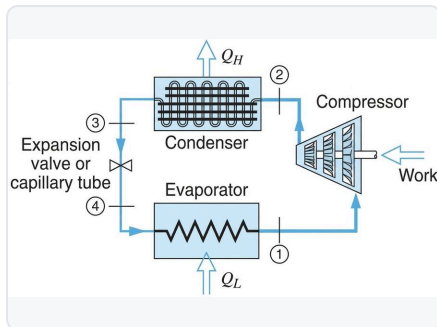
- Confirmed **<6%** deviation between COMSOL and analytical solution
- Identified **René 41** as optimal material for shaft thermal protection
- Demonstrated mesh-independent convergence

Automotive HVAC Refrigerant Cycle Simulation

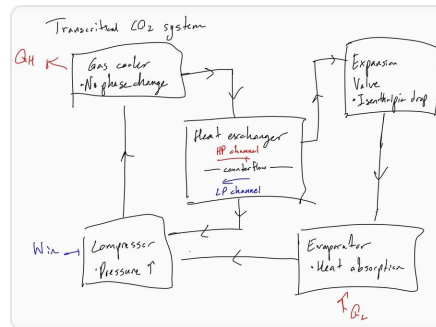
Thermodynamics

Python

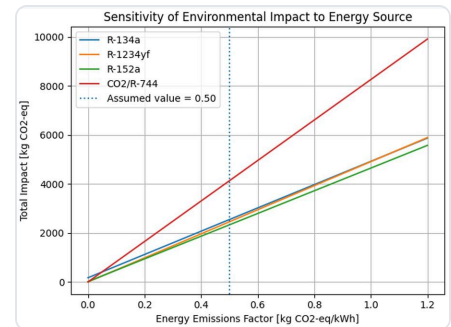
TEWI Analysis



CYCLE DIAGRAM



COP OPTIMIZATION



TEWI RESULTS

WHAT?

- Modeled four automotive refrigerants across conventional and transcritical vapor-compression cycles in Python
- Investigated lifecycle environmental impact of refrigerant selection

HOW?

- Built simulation in **Python** covering conventional and transcritical vapor-compression cycles
- Optimized CO₂/R-744 transcritical high-side pressure to maximize COP
- Balanced cooling capacity against compressor work

RESULTS

- Conducted TEWI analysis revealing **indirect emissions dominate** climate impact
- Found near-zero GWP insufficient to minimize total environmental footprint

PROJECT

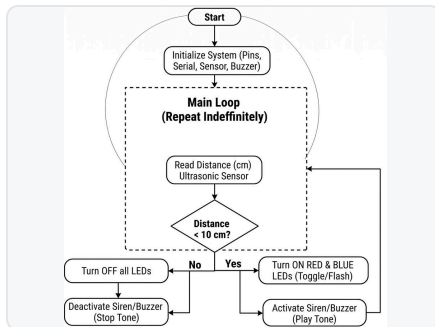
Autonomous Ambulance

Robotics

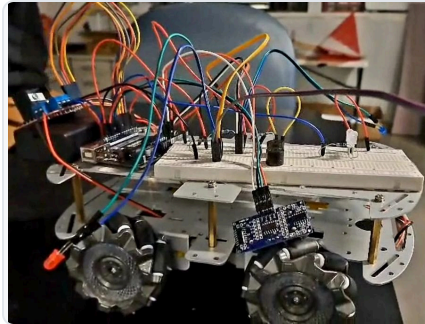
Arduino

C++

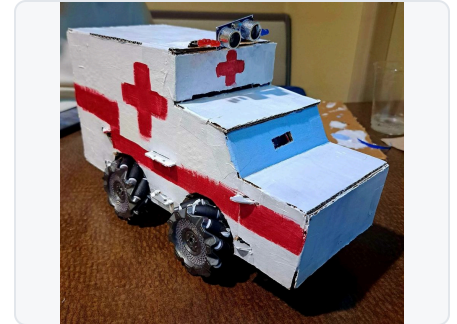
SolidWorks



SYSTEM FLOWCHART



CIRCUITRY



FINISHED VEHICLE

WHAT?

- Developed autonomous driving system with collision avoidance, adaptive siren logic, and patient transport ability
- Focused on reducing vehicle weight and project overhead

HOW?

- Modeled chassis in **SolidWorks**
- Developed **C++** control firmware for **Arduino** microcontroller and implemented associated circuitry
- Integrated **ultrasonic (sonar) sensors** for autonomous navigation and obstacle avoidance
- Implemented functional emergency lights and sirens

RESULTS

- Achieved full vehicle autonomy
- Demonstrated **100%** obstacle avoidance success
- Validated patient transport capabilities
- Reduced chassis weight by **50%** via design optimization
- Completed project **30%** under budget

PROJECT

Flood Disaster Model

Army Corps of Engineers

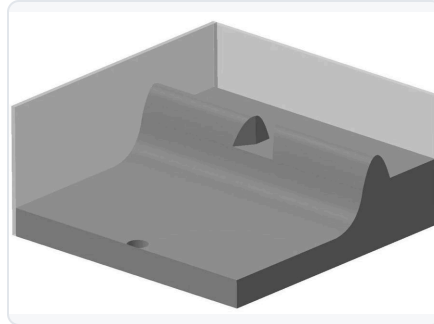
CFD

SolidWorks

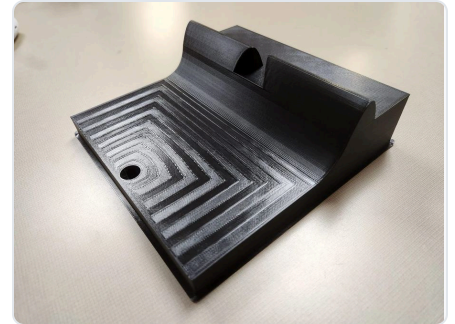
3D Printing



CFD SIMULATION



CAD MODEL



PETG PROTOTYPE

WHAT?

- Designed model to simulate water propagation following a catastrophic levee failure
- Focused on creating model that accurately replicates fluid behavior

HOW?

- Used **SolidWorks CFD** to optimize design for maximum flow realism
- Sliced and printed final design via **3DprinterOS**

RESULTS

- Produced **3D-printed PETG** prototype for physical validation and testing
- Delivered validated design and fabrication-ready files to subsequent development team

Skills

Technical

SolidWorks

Ansys

MATLAB

Python

Java

COMSOL

XFLR5

Arduino

Excel

Methods

Computational Fluid Dynamics

Finite Element Analysis

AI-Assisted Workflows

GD&T

Equipment

CNC

Lathe

Mill

Drill Press

Laser Cutter

3D Printing

Band Saw

Soldering

Brazing

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